



BITS Pilani presentation

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Pilani | Dubai | Goa | Hyderabad

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SS ZG622

Software Project Management

Lecture No. 6

Recap - Effort estimation



- Function point estimation in plan-driven process models
- Story points estimation in agile process models
- Horizontal vs vertical slicing of stories
- Differences between estimation approaches

Project Scheduling - Topics



- Plan-driven methods
 - Mapping effort to duration
 - Examples of effort and time distribution
 - Work breakdown structure (WBS)
 - Gantt charts
 - AoN Diagrams and critical path
- Agile methods
 - Example of effort distribution in agile methods
 - Example planning tool - Pivotal Tracker (user stories, bugs, chores and milestones)

Scheduling in plan-driven methods



- Defining project tasks and milestones
- Mapping estimated effort to duration
- Identifying dependencies
- Developing a work breakdown structure (WBS)
- Creating gantt charts or AoN diagrams
- Finding critical path(s)
- *Monitoring and updating schedule* (as part of execution phase)

Mapping estimated effort to duration



- Effort in function points or KLOC
- Effort to duration
 - Experience-based and heuristics-based methods
 - Parametric methods like COCOMO81 and COCOMO II



W06-#1 Suggested project duration

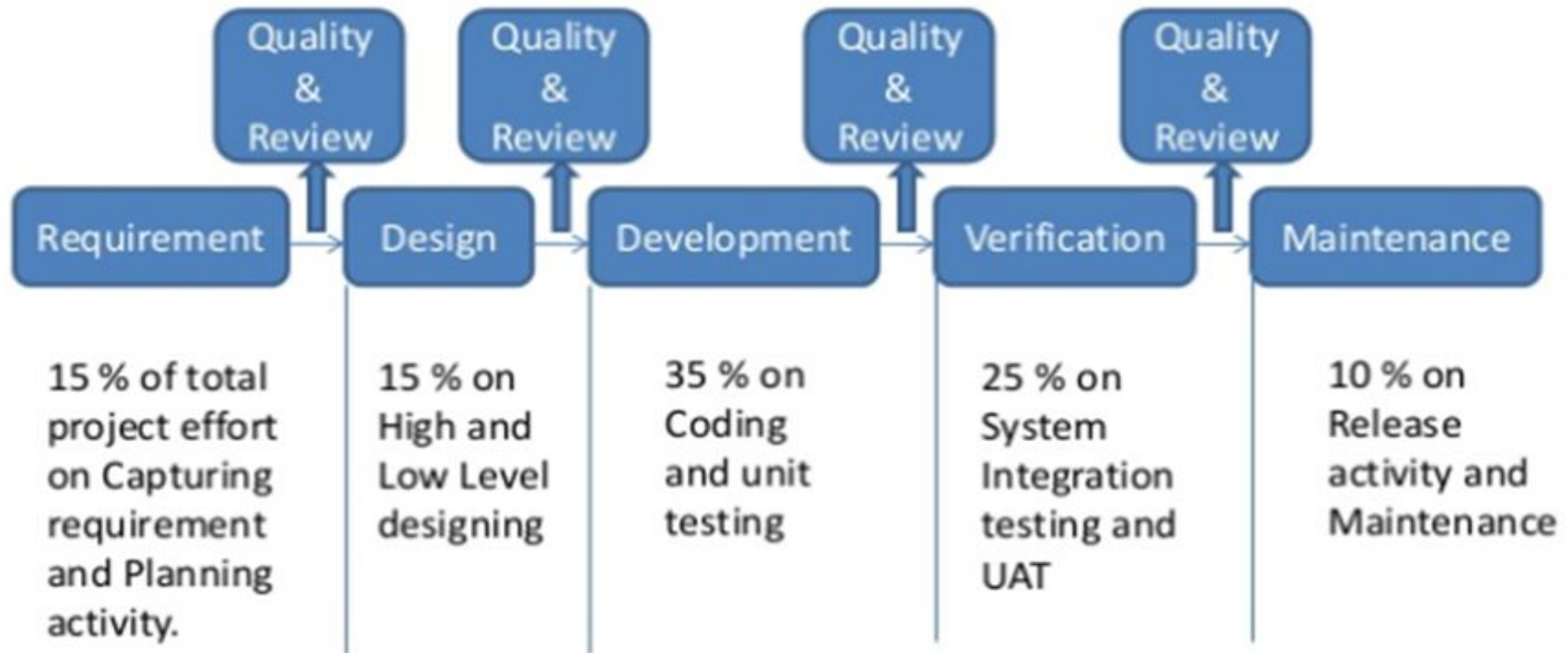
Assuming that the effort and productivity for the Mux-core project are as follows:

- Function points ~ 500
- Average productivity 10 function points per month per developer
- # of developers available is flexible (between 1 to 20)

What should be the ideal project duration?

<https://www.ifpug.org/wp-content/uploads/2017/04/IYSM.-Thirty-years-of-IFPUG.-Software-Economics-and-Function-Point-Metrics-Capers-Jones.pdf>

Example of effort distribution



<https://www.slideshare.net/brickedestimation/effort-distributiononwaterfallandagile>

Example of time distribution



Phase	Tasks	% of Total Time
Requirements Gathering & Analysis	Requirements documentation, stakeholder meetings	10%
System Design	High-level design, detailed design	15%
Implementation (Coding)	Module coding, unit testing	35%
Integration & Testing	System testing, bug fixing	25%
Deployment	Deployment setup, production roll-out	10%
Buffer/Contingency	Risk mitigation, buffer for delays	5%

COCOMO formula for duration



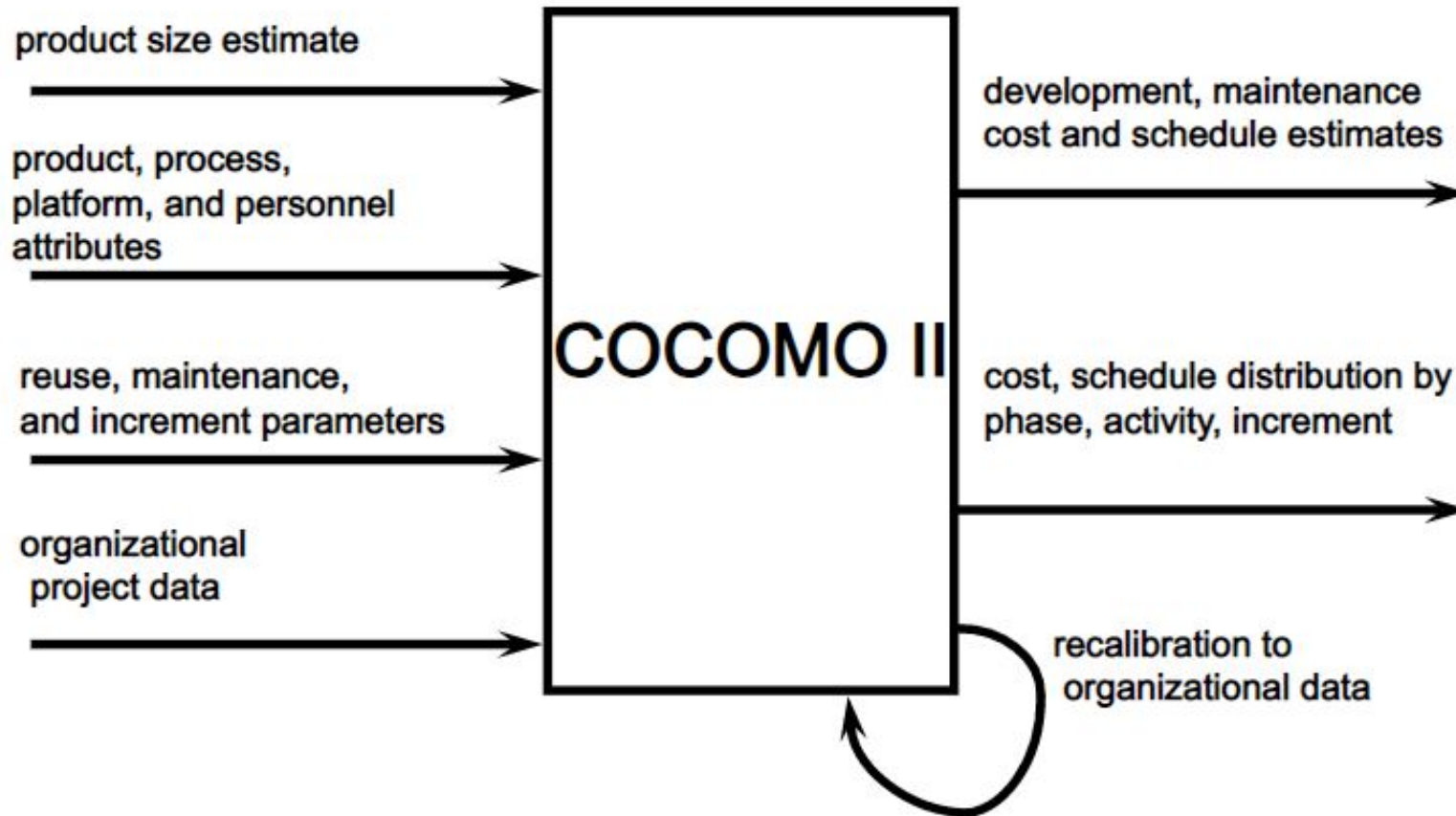
COCOMO formula for duration calculation: $M = a \times E^b$
where E is the estimated effort and

Project Type	a	b
Organic	2.5	0.38
Semi-detached	2.5	0.35
Embedded	2.5	0.32

Assuming that Mux-Core is a semi-detached project type with 500 fp, the project duration should be 22 months

See <https://www.geeksforgeeks.org/software-engineering/software-engineering-cocomo-model/> for details

COCOMO Black Box Model



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Brief introduction -

<https://www.geeksforgeeks.org/software-engineering-cocomo-ii-model/>

<https://slideplayer.com/slide/7947860/>

COCOMO II Calculator (read chap 5 of the text book for details)



Software Size Sizing Method **Function Points** ▼

Unadjusted Function Points **440** Language **Java** ▼

Software Scale Drivers

Precedentedness	Nominal ▼	Architecture / Risk Resolution	Nominal ▼	Process Maturity	Nominal ▼
Development Flexibility	Nominal ▼	Team Cohesion	Nominal ▼		

Software Cost Drivers

Product		Personnel		Platform	
Required Software Reliability	Nominal ▼	Analyst Capability	Nominal ▼	Time Constraint	Nominal ▼
Data Base Size	Nominal ▼	Programmer Capability	Nominal ▼	Storage Constraint	Nominal ▼
Product Complexity	Nominal ▼	Personnel Continuity	Nominal ▼	Platform Volatility	Nominal ▼
Developed for Reusability	Nominal ▼	Application Experience	Nominal ▼	Project	
Documentation Match to Lifecycle Needs	Nominal ▼	Platform Experience	Nominal ▼	Use of Software Tools	Nominal ▼
		Language and Toolset Experience	Nominal ▼	Multisite Development	Nominal ▼
				Required Development Schedule	Nominal ▼

Maintenance **Off** ▼

<http://softwarecost.org/tools/COCOMO/>

COCOMO II Calculator (read chap 5 of the text book for more details)

innovate

achieve

lead

Results

Software Development (Elaboration and Construction)

Effort = 169.9 Person-months

Schedule = 20.0 Months

Cost = \$4246906

Total Equivalent Size = 40000 SLOC

Effort Adjustment Factor (EAF) = 1.00

12 169.9 20.0 4246906 40000 1.00

13

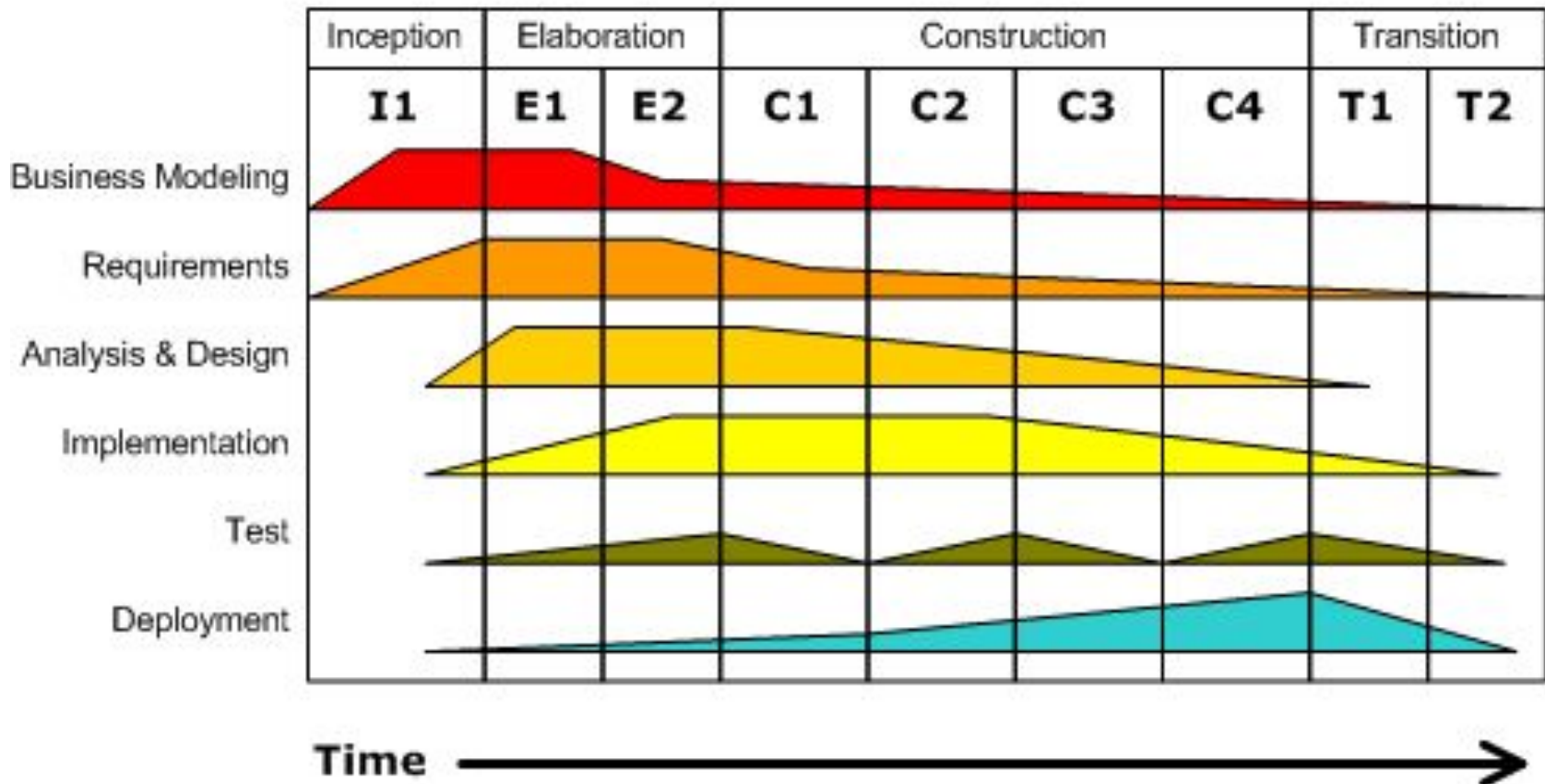
Note: Assume \$ = INR for interpretation of the inputs and results;; i.e., 25k rupees average salary per month

Unified Process (for explaining estimation results)



Iterative Development

Business value is delivered incrementally in time-boxed cross-discipline iterations.



COCOMO II Calculator



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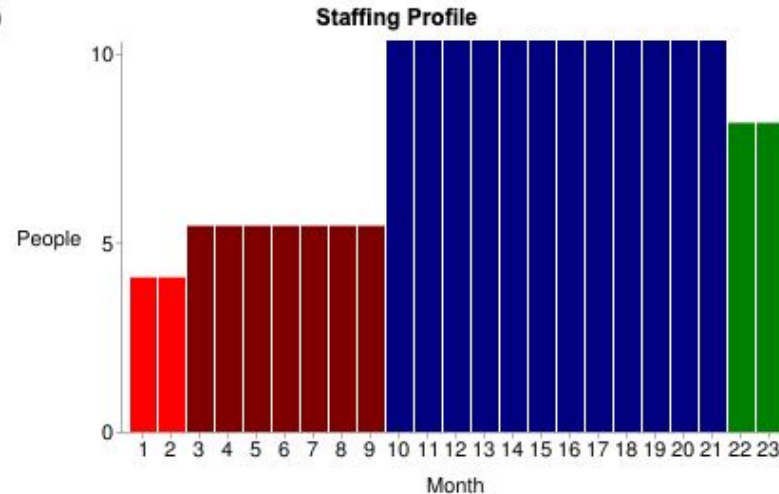
Cost = \$4246906

Total Equivalent Size = 40000 SLOC

Effort Adjustment Factor (EAF) = 1.00

Acquisition Phase Distribution

Phase	Effort (Person-months)	Schedule (Months)	Average Staff	Cost (Dollars)
Inception	10.2	2.5	4.1	\$254814
Elaboration	40.8	7.5	5.4	\$1019258
Construction	129.1	12.5	10.3	\$3227649
Transition	20.4	2.5	8.2	\$509629



Software Effort Distribution for RUP/MBASE (Person-Months)

Phase/Activity	Inception	Elaboration	Construction	Transition
Management	1.4	4.9	12.9	2.9
Environment/CM	1.0	3.3	6.5	1.0
Requirements	3.9	7.3	10.3	0.8
Design	1.9	14.7	20.7	0.8
Implementation	0.8	5.3	43.9	3.9
Assessment	0.8	4.1	31.0	4.9
Deployment	0.3	1.2	3.9	6.1

<http://softwarecost.org/tools/COCOMO/>

COCOMO II Calculator (read chap 5 of the text book for details)



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Note: Assume \$ = INR for interpretation of the inputs and results;; i.e., 25k rupees average salary per month

W06-#2 COCOMO II Calculator

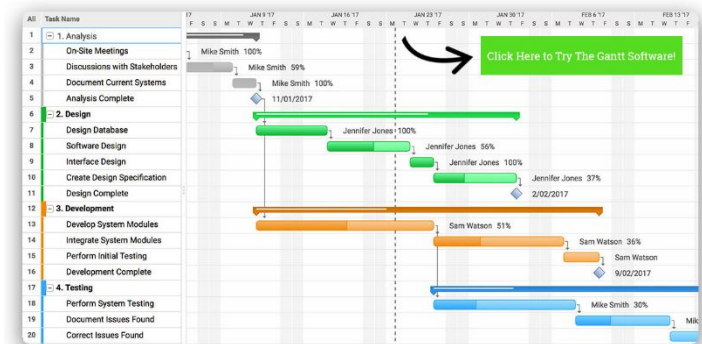


Keeping the function points as 500, observe the effect of software scale and cost drivers on the estimated effort, schedule and cost.

<http://softwarecost.org/tools/COCOMO/>

WBS and Project Plan

- **Work Breakdown Structure (WBS):** Breaking down the work into parts and assign these to project team members, anticipate problems that might arise and prepare tentative solutions to those problems.
- **Project Plan**, which is created at the start of a project, is used to communicate how the work will be done to the project team and customers, and to help assess progress on the project.



Example of a WBS fragment



3 Physical Design

3.01 Design or specify physical database

- 3.01.01 Review logical database design
- 3.01.02 Determine access methods to be used
- 3.01.03 Normalize database
- 3.01.04 Design database architecture
- 3.01.05 Identify reusable database structures
- 3.01.06 Develop detailed database layout
- 3.01.07 Develop database file, record, and schema descriptions
- 3.01.08 Develop module calling sequences
- 3.01.09 Update data dictionary entries
- 3.01.10 Validate physical database design

Example of a WBS fragment

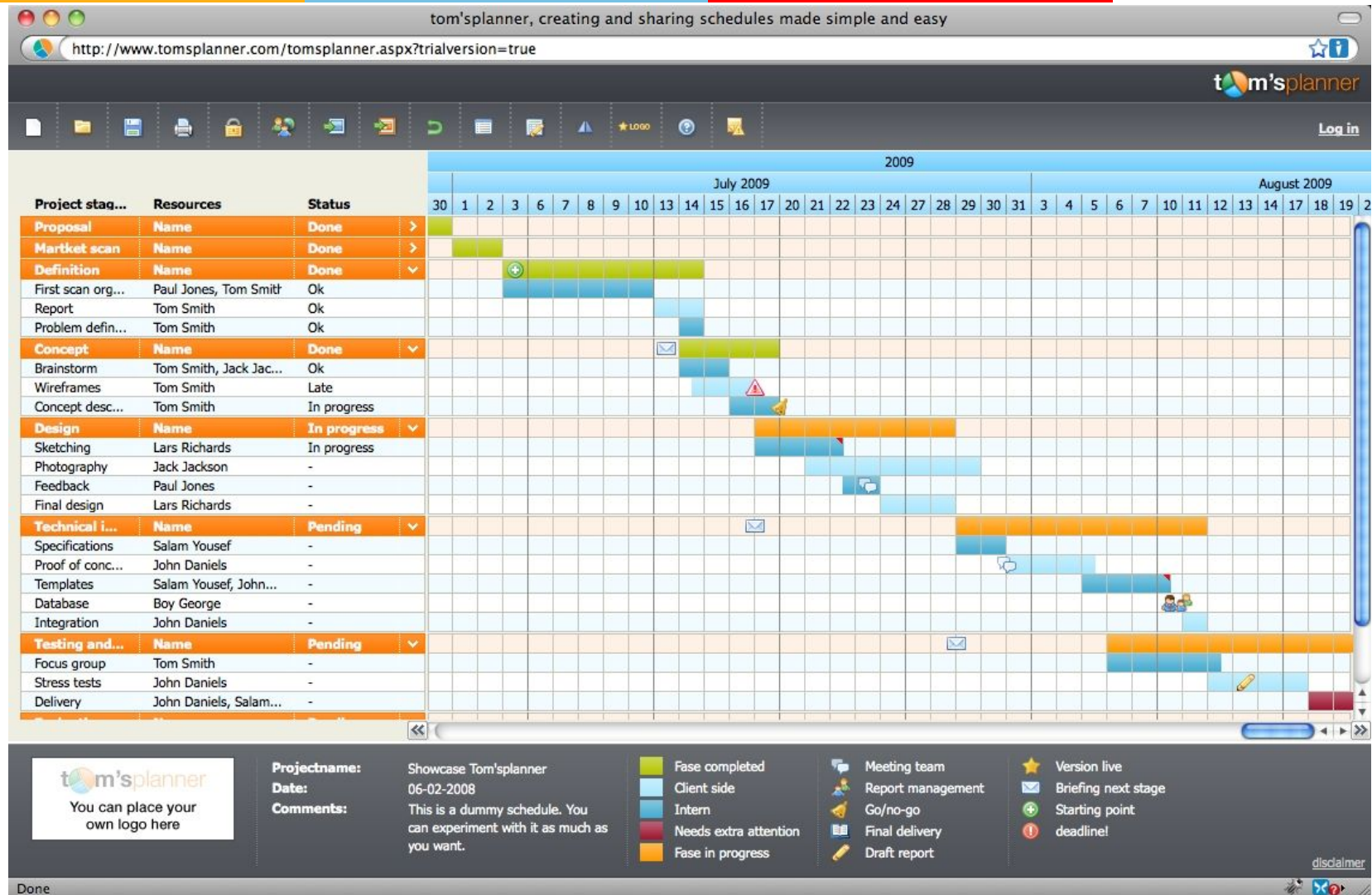


Level 3: Sub-Tasks within Each Phase

1. Requirements Gathering and Analysis

- **1.1. Requirement Elicitation**
 - Stakeholder interviews
 - Workshops
 - Surveys and questionnaires
 - Competitor analysis
- **1.2. Requirement Documentation**
 - Functional Requirements Specification (FRS)
 - Non-Functional Requirements Specification (NFRS)
- **1.3. Requirement Review and Validation**
 - Requirements review sessions
 - Sign-off by stakeholders
- **1.4. Feasibility Study**
 - Technical feasibility
 - Budget and resource feasibility

Activity planning – Gantt Charting tool



W06-#3 Gantt Chart

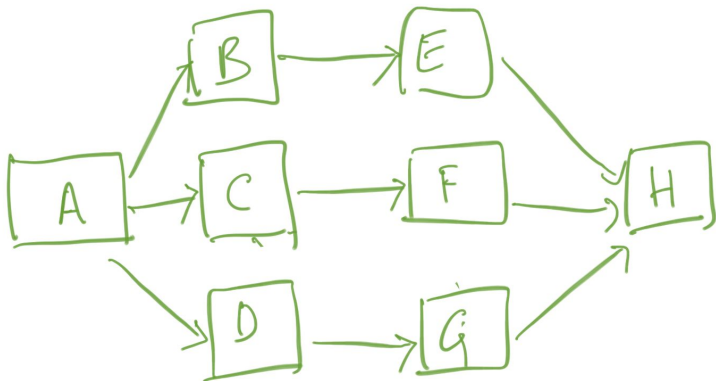


Activity	Duration (days)	Depends on	Resource
A	3		SA
B	1	A	SD1
C	2	A	SD2
D	4	A	SD2
E	3	B	CD1
F	3	C	CD1
G	6	D	CD2
H	3	E,F,G	SA

Prepare a Gantt Chart (on paper or any tool you can find) to find the critical path and project duration

Activity on Node Diagrams

Early Start	Duration	Early Finish
Task Name		
Late Start	Slack	Late Finish

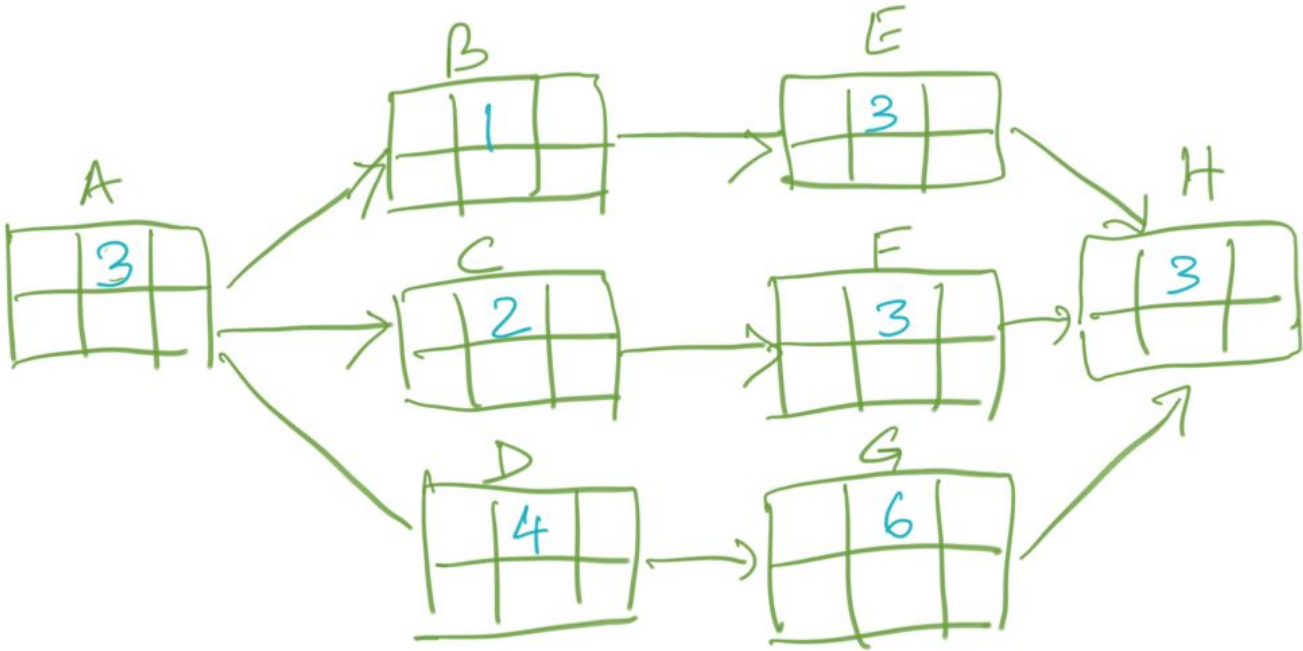


Activity	Duration (days)	Depends on
A	3	
B	1	A
C	2	A
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H	3	E,F,G

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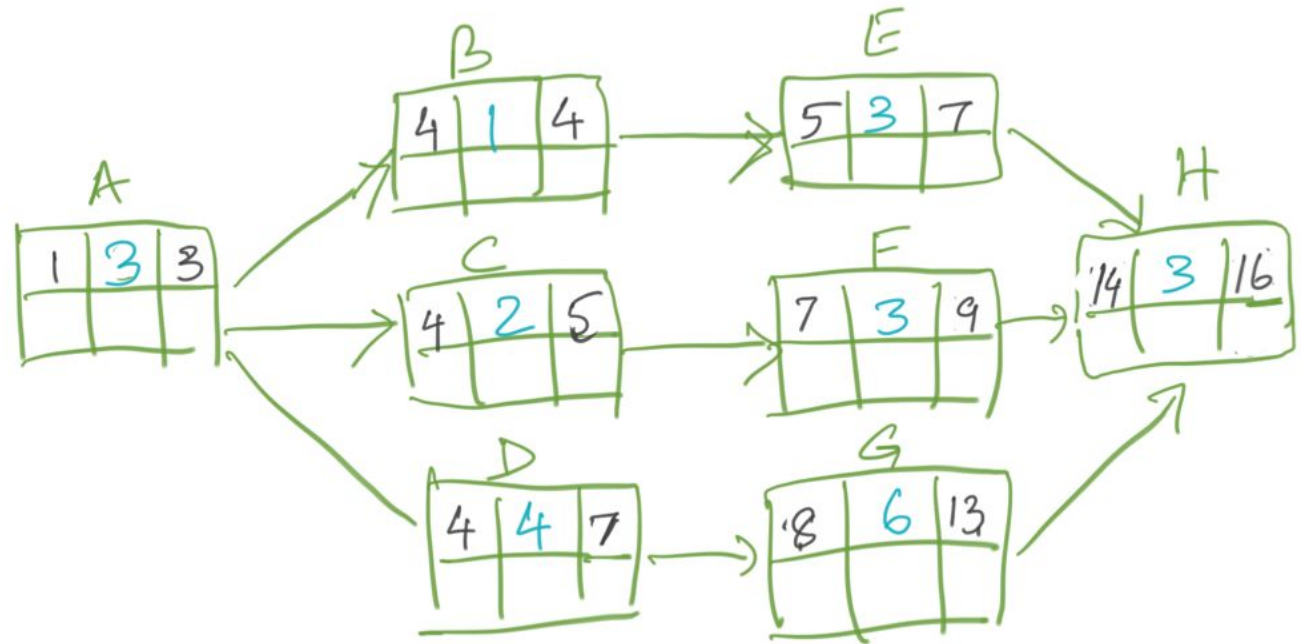
Forward pass in Activity on Node Diagrams



When an activity (e.g., H) is dependent on multiple activities then take the max of EFs

Early Start	Duration	Early Finish
Task Name		
Late Start	Slack	Late Finish

Activity	Duration (days)	Depend s on
A	3	
B	1	A
C	2	A
D	4	A
E	3	B
F	3	C
G	6	D
H	3	E,F,G



$$\text{Early Finish (EF)} = \text{Early Start (ES)} + \text{Duration} - 1$$

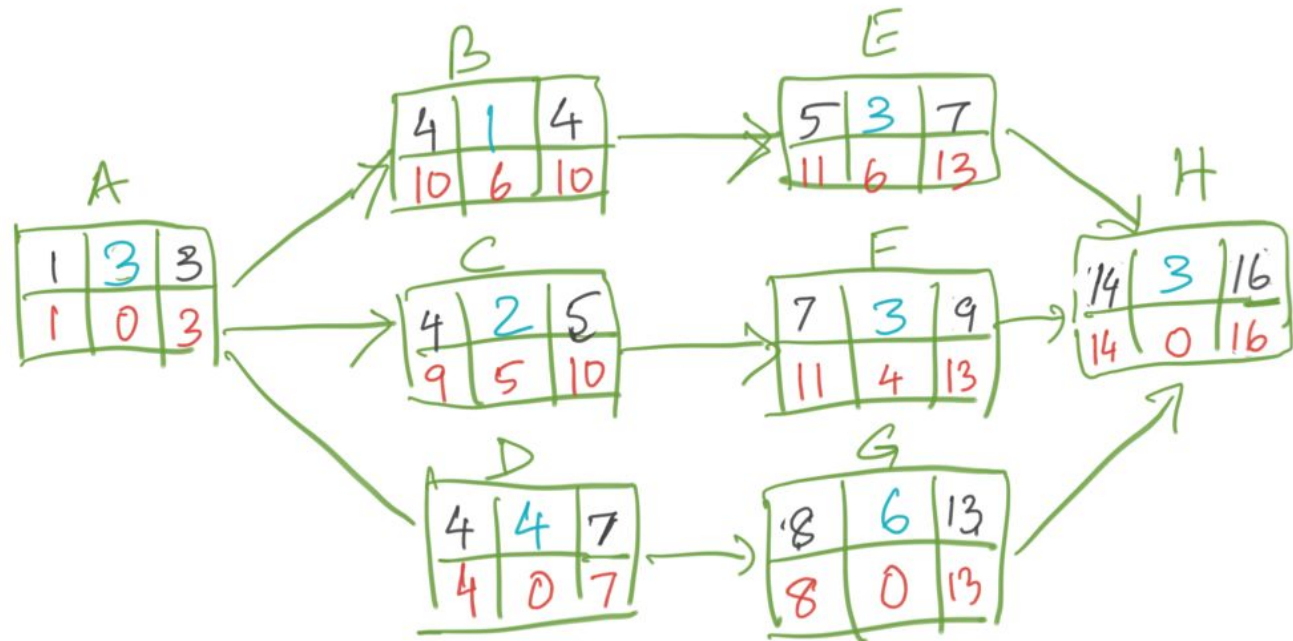
Backward pass in Activity on Node Diagrams



When an activity (e.g., A) is branching to multiple paths then take the min of LSs

Early Start	Duration	Early Finish
Task Name		
Late Start	Slack	Late Finish

Activity	Duration (days)	Depend s on
A	3	
B	1	A
C	2	A
D	4	A
E	3	B
F	3	C
G	6	D
H	3	E,F,G



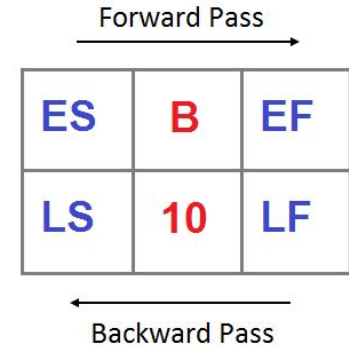
$$LS = LF - Duration + 1$$

$$Float/Slack = LS - ES \text{ or } LF - EF$$

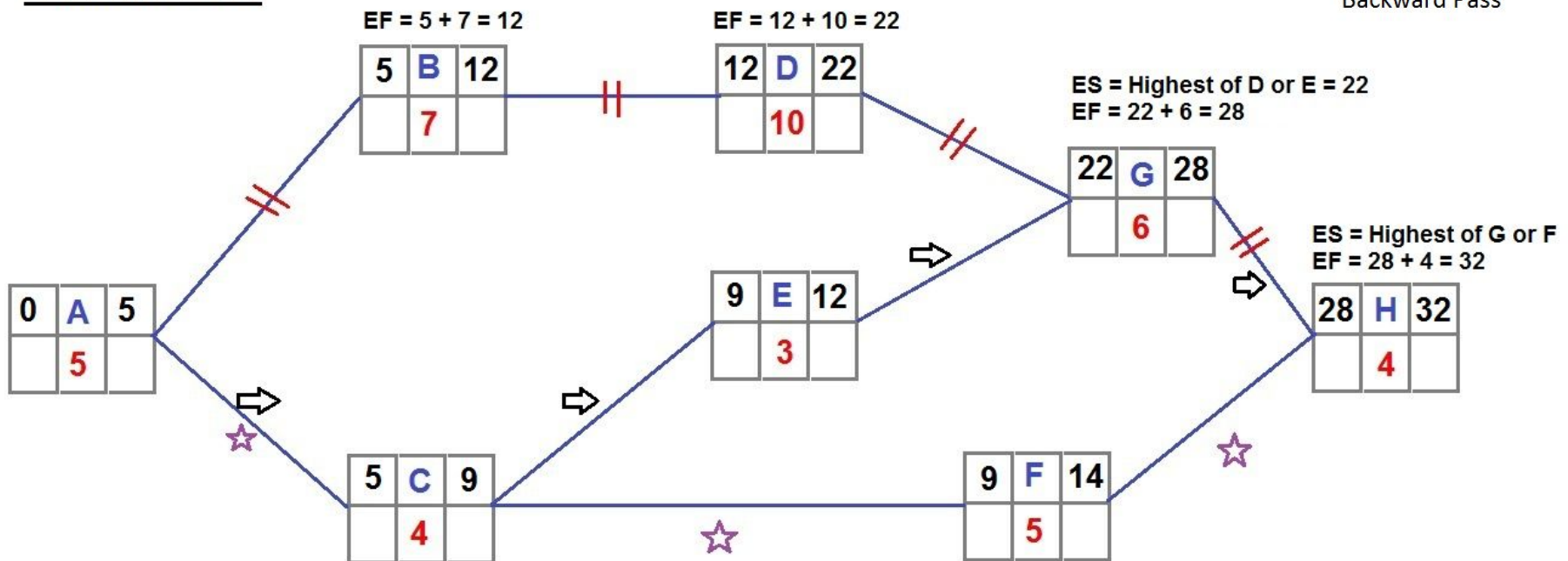
Activity on Node Diagrams (with 0 start instead of 1)



$$\text{Early Finish (EF)} = \text{Early Start (ES)} + \text{Duration}$$



FORWARD PASS



<https://tiemchart.com/>

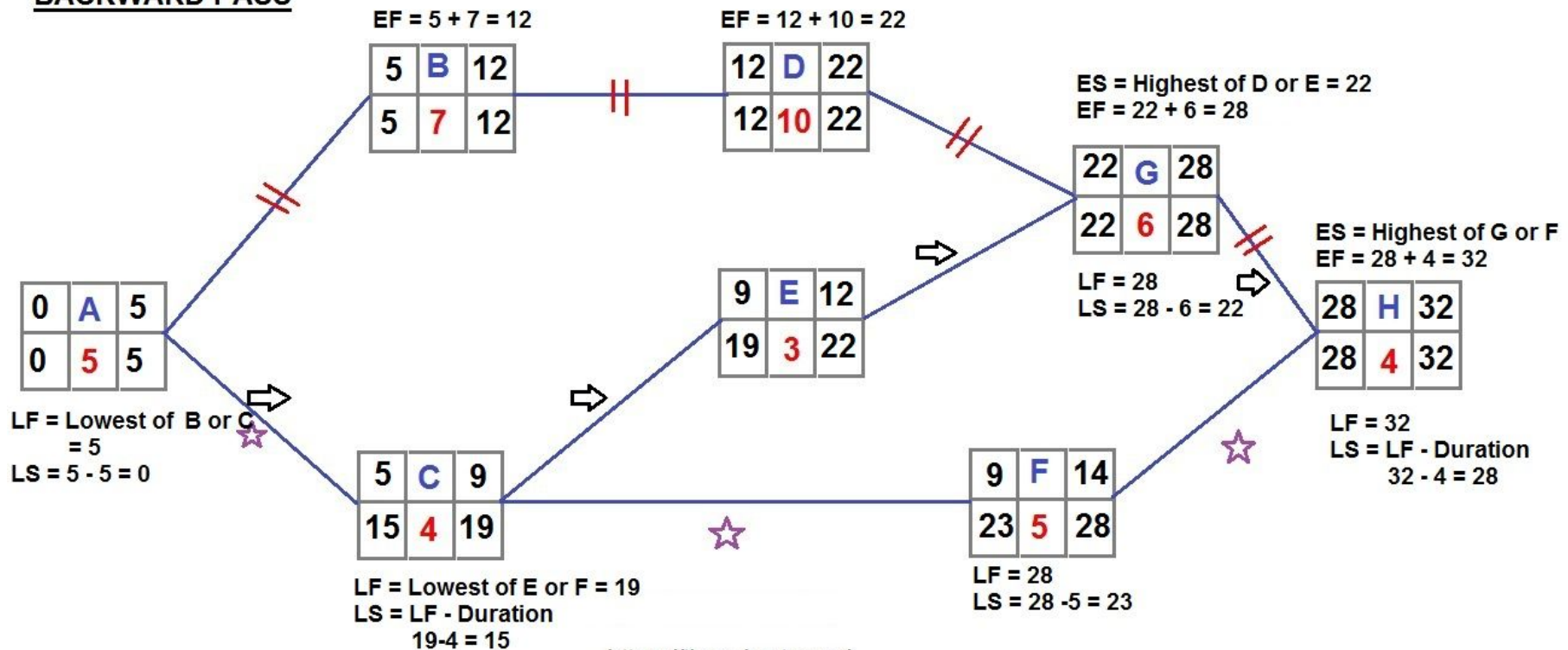
<https://tiemchart.com/blogs/training/forward-pass-and-backward-pass/>

Activity on Node Diagrams (with 0 start instead of 1)



$LS = LF - \text{Duration}$ (float/slack is not shown in the diagram)

BACKWARD PASS



<https://tiemchart.com/>

<https://tiemchart.com/blogs/training/forward-pass-and-backward-pass/>

AoN Summary



- Forward Pass gives us the earliest start (ES) and earliest finish (EF) of each activity.
- Backward Pass gives us the latest start (LS) and latest finish (LF).
- Float/slack is calculated by comparing the results of forward and backward passes, indicating how much flexibility there is for each activity.
- The critical path is the sequence of activities with zero float, representing the longest path through the project



Gantt chart vs AoN diagram

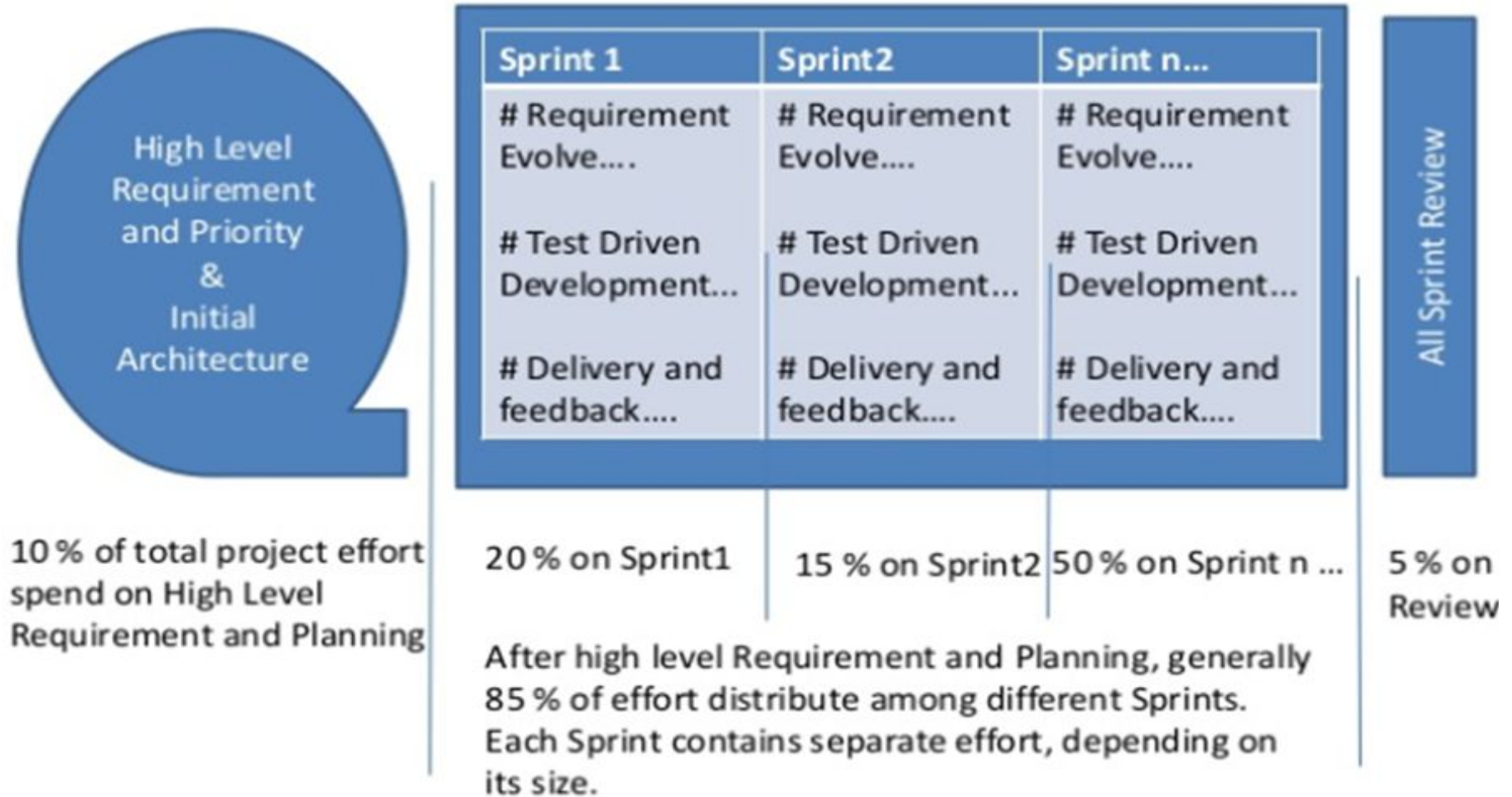
Compare suitability of Gantt charts and
AoN diagrams for ACL Mux Core project scheduling

Scheduling in agile methods



- Allocate time for **sprint 0** activities
- Distribution of effort and time (not uniform across sprints)
- Example agile project management tool

Example of effort distribution in agile process (Scrum)



<https://www.slideshare.net/brickedestimation/effort-distributiononwaterfallandagile>

WBS in agile projects

- High-level planning with WBS (Epics and release planning)
- Product -> Epics -> User stories -> Tasks
- Dynamic WBS
- Time-boxed WBS

Pivotal Tracker(example)

https://www.pivotaltracker.com/help/articles/creating_a_project/



Pivotal Tracker (example)



- Epics
- Icebox
- My work
- Story types
 - Features
 - Chores
 - Bugs
 - Releases

W06-#5 Gantt chart and AoN diagram in agile projects



Do you think Gantt charts and AoN diagrams are relevant for ACL Mux Core project to be executed using Scrum framework ? Explain.